

Carbon Pricing Policies and Global Climate Change – Critical Perspectives

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Abstract

The paper considers carbon pricing policies that have developed in response to global climate change concerns. There have been significant international institutional climate change policy drivers to harness a global effort to reduce adverse anthropogenic (human-induced) effects on the climate system. Following the Kyoto Protocol (1997) and then the Paris Agreement (2015) many countries have adopted carbon pricing policies with various mechanisms that in combination had achieved coverage of some 28 per cent. of global carbon emissions by 2025 according to the World Bank. There are both mandatory and voluntary carbon markets that are effectively specialised human-construct commodity markets. Carbon pricing has encouraged structured financial engineering based on measured carbon emissions in different jurisdictions. Further research will be necessary to assess whether the progress in carbon pricing and carbon markets is sufficiently effective in achieving positive outcomes at national and global levels. Governments may face increasing pressure to consider the balance between carbon pricing policies and promotion of carbon management technologies especially if the latter prove to be more effective and/or acceptable to businesses and the wider public. This paper identifies some controversies and challenges in the development of carbon pricing and concludes with a consideration of important issues to address in the future.

I Introduction – From the Science to the Politics of Climate Change

This paper will provide critical perspectives on carbon pricing in the UK. Carbon dioxide (CO₂) emissions pricing or the shorter more convenient form ‘carbon pricing’ (CP) that is frequently used as in this paper, refers to mechanisms such as carbon taxes, emissions trading systems (ETSs) or ‘cap-and-trade systems’ or various hybrid systems devised for tackling climate change (Kirkpatrick, 2022). The CP systems that are operating in a number of countries in the world have been introduced after many years of scientific developments to produce and analyse evidence and significant international political initiatives. Awareness and discussion of CO₂ as one particular ‘greenhouse gas’, has increased significantly in the Twenty First Century because of concerns about the effects of human activity-related CO₂ emissions on climate change. It took a long time for science to influence government policy and then for that policy to be reflected in CP mechanisms.

Scientific studies on what became known as the ‘greenhouse effect’ can be traced back to the nineteenth century, for example in papers by Joseph Fourier who proposed the hypothesis that gases in the earth’s atmosphere must create heat-trapping barriers (Fourier, 1824 and Fourier, 1837) and Eunice Newton Foote who performed experimental work in 1856 (Foote 1856 and Shapiro, 2021) then John Tyndall produced precision equipment for experiments between 1859 and 1861 with findings that showed the powerful heat trapping effect of CO₂ (Tyndall, 1861), and then there were later contributions such as those by Svante Arrhenius in 1896 who created perhaps the first model of climate change and predicted that a doubling of the amount of CO₂ in the Earth’s atmosphere would increase the Earth’s temperature by between 5 and 6 degrees Centigrade (Arrhenius et al., 2008). The estimations of the rate of change of global warming led to significant upward revisions of the greenhouse effect as more data was produced and measurement improved especially in the latter part of the Twentieth Century.

If the Nineteenth Century research provided the theory and models for analysis of greenhouse gases and heat-trapping effects on climate change, then the Twentieth Century was significant for the considerable improvements in data collection and measurement of the levels of CO₂ emissions as well as more accurate data on changes in global temperatures. The 1960s and 1970s witnessed the development of satellite technology such as NASA’s Nimbus satellite that provided global atmospheric temperature measurements (NASA, 2025). In the 1970s with the increasing media attention responding to increased availability of data on a variety of environmental issues - particularly pollution although climate change was also on the menu - there was increased activism with rallies and demonstrations in some cases, for example the first Earth Day took place in 1970 (Earth Day, 2025).

The 1980s and 1990s saw the rapid advancement of computer technology with advanced software helping to produce more powerful models for analysis and predictions of climate change effects. In the first quarter of the Twenty First Century the scientific evidence has continued to grow with more critical analysis of aspects such as regional variations. The development of the international politics of climate change can also be traced back to the 1970s with significant involvement by organisations such as the United Nations and the

World Meteorological Organization, and this will be discussed in terms of the policy drivers for carbon pricing in a number of countries such as the UK.

II International Institutional Climate Change Policy Drivers

The United Nations provided the forum for different nations to discuss the global environment in June 1972 when it held the UN Conference on the Human Environment (UNCHE) in Stockholm which concluded by identifying the need for states to work cooperatively to solve environmental issues on a global scale (United Nations, 2025a). The World Meteorological Organization (WMO) organised the First World Climate Conference in 1979 and this called for the urgent development of “a common strategy for a greater understanding of the climate system and a rational use of climate information” and there was also the proposal to establish the World Climate Programme (WCP) that apply understanding of the climate system “for the benefit of societies coping with climate variability and change” (World Meteorological Organization, 2025 - online).

The WMO and the United Nations Environment Programme established the Intergovernmental Panel on Climate Change (IPCC) in 1988 (IPCC, 2025a) with an initial task that was outlined in UN General Assembly Resolution 43/53 of 6 December 1988 (IPCC, 2025b – page 134) as follows:

“to prepare a comprehensive review and recommendations with respect to:

- (a) the state of knowledge of the science of climate change;
- (b)the social and economic impact of climate change including global warming,
- (c) possible response strategies to delay, limit or mitigate the impact of adverse climate change;
- (d) the identification and possible strengthening of relevant existing international legal instruments having a bearing on climate (and)
- (e) elements for inclusion in a possible future international convention on climate.”

The decade of the 1990s witnessed major developments in global attitudes to climate change concerns. The United Nations Framework Convention on Climate Change (UNFCCC) is a multilateral treaty adopted in 1992 and entered into force in 1994 with the purpose of stabilizing greenhouse gas concentrations “at a level that would prevent dangerous anthropogenic (human-induced) interference with the climate system” (United Nations, 2025a, online). The UNFCCC is a framework that has provided the basis for international climate negotiations including the landmark agreements such as the Kyoto Protocol (1997) and the Paris Agreement (2015) that have been drivers of climate change policy in many of the world’s countries including the UK.

There are currently 198 Parties to the UNFCCC (United Nations, 2025b), these are known as ‘Parties to the Convention’, and there is an annual Conference of the Parties (COP) with the stated objectives of measuring progress and negotiating multilateral responses to climate change. The first COP was held in Berlin, Germany in 1995 and at the time of writing the most recent COP was COP29 that was held in Baku, Azerbaijan, in November 2024 – a historical summary of the COPs is shown in Figure A in the Appendix. There have been major challenges over the thirty years of COPs in achieving consensus on more than broad principles and in some years, such as COP 6 in 2000 and COP 18 in 2012 there have been

disagreements between significant parties. As the Kyoto Protocol (1997) that emerged from COP 3 in 1997 and the Paris Agreement (2015) that emerged from are particularly relevant to the UK carbon pricing system and the way it has developed both before and after Brexit, these will be considered in this paper in the next sections.

III The Kyoto Protocol (1997) and the Kyoto Mechanisms

The Kyoto Protocol was an outcome of COP 3 held in Kyoto, Japan and was concluded On 11th December 1997 (Kyoto Protocol, 1997). The Kyoto Protocol was introduced to bring the UNFCCC into operation by committing industrialized countries and economies in transition to limit and reduce greenhouse gas emissions according to their agreed individual national targets – a peculiar feature is that although signed in December 1997, the Kyoto Protocol only entered force in February 2005 due to a complex and lengthy ratification process and there were believed to be 192 Parties by 2000 although it has now been superseded by the Paris Agreement (2015).

Although the Kyoto Protocol (1997) has been superseded, the Kyoto Mechanisms that were introduced under the Kyoto Protocol remain relevant as they played a significant part in the creation of the carbon market. The Kyoto Mechanisms were introduced to achieve three objectives (UNFCCC, 2025a):

1. to stimulate sustainable development through technology transfer and investment
2. to help countries with Kyoto commitments to meet their targets by reducing emissions or removing carbon from the atmosphere in other countries in a cost-effective way
3. to encourage the private sector and developing countries to carry out emission reduction or removal enhancement projects in other developed countries.

There are three Kyoto Mechanisms (UNFCCC, 2025a):

1. Clean Development Mechanism (CDM) – this is defined in Article 12 of the Kyoto Protocol and allows a country with an ‘emission-reduction’ or ‘emission-limitation’ to implement an emission-reduction in developing countries to earn saleable Certified Emission reduction (CER) credits, each equivalent to one tonne of CO₂ which can count towards Kyoto targets.
2. Joint Implementation (JI) – this is defined in Article 6 of the Kyoto Protocol allows a country with an ‘emission-reduction’ or ‘emission-limitation’ commitment under the Protocol (Annex B Party) to earn emission reduction units (ERUs) from an emission-reduction or emission removal project in in another Annex B party, each equivalent to one tonne of CO₂ which can count towards Kyoto targets.
3. Emissions trading (ET) – Parties with commitments under the Kyoto Protocol (Annex B Parties) accepted targets for limiting or reducing emissions. Emissions trading as set out in Article 17 of the Kyoto Protocol allows countries that have emission units to spare, that is, units permitted but not used, to sell the excess capacity to countries that are over their targets. In this way a new commodity was created in the form of emission reductions or removals. A ‘carbon market’ has been born as carbon is now traded like any other commodity.

The Kyoto Mechanisms still have some relevance even though the Kyoto Protocol (1997) has now been superseded by the Paris Agreement (2015) that will now be discussed in the next section. Ten years of international discussions following the Kyoto Protocol (1997) commitments brought responses from a number of industrialized countries including the UK which at the time was still a member of the EU and by 2008 was to introduce a historic piece of legislation in the form of the Climate Change Act establishing one of the earliest examples of a legally binding 'Carbon Target' (Climate Change Act, 2008).

IV The Paris Agreement (2015)

The Paris Agreement (United Nations, 2015) was signed by Parties at COP 21 in Paris on 12 December 2015, coming into force on 4 November 2016 as a legally binding international treaty on climate change (UNFCCC, 2025b). The date of 4 November 2016 is a precise date determined as "...thirty days after the date on which 55 Parties to the Convention accounting in total for at least an estimated 55% of the total global greenhouse gas emissions have deposited their instruments of ratification, acceptance approval or accession with the Depositary" (UNFCCC, 2025c). At the time of writing, 195 Parties out of 198 Parties to the UNFCCC are Parties to the Paris Agreement. There are a number of key terms of the Paris Agreement and obligations on the signatory nations:

1. Global temperature increase is to be limited to well below 2 Degrees Celsius and at the same time efforts are to be pursued to limit global temperature increase to 1.5 Degrees Celsius above pre-industrial levels.
2. Each nation is required to determine and submit its own climate action plan in the form of Nationally Determined Contributions (NDCs) set its own emission reduction targets that are reviewed every five years to raise ambitions.
3. Richer nations are expected to help poorer nations with climate finance to adapt to climate change and transition to renewable energy. Climate finance is presented as a crucial part of building resilience and reducing the vulnerability of poorer nations.

There are also other significant features of the Paris Agreement, for instance:

1. Technology transfer, development and cooperation is to be encouraged.
2. Transparency is a fundamental feature to allow progress and accountability to be tracked.

The US is the most significant nation in a very short list of nations that are not Parties to the Paris Agreement (at the time of writing). The US officially joined the Paris Agreement in September 2016 but then officially withdrew from the Agreement in 2020 during the administration of President Trump. The US then re-joined the Agreement in 2021 during the administration of President Biden. More recently, following another change of administration, the US White House announced on 20 January 2025 that it would withdraw from the Paris Agreement (The White House, 2025).

A key part of the ongoing global discussions is the collection of data and the analysis of available information to determine both the levels of and changes in the greenhouse gas (GHG) emissions at national and global level. Many organisations are involved in the collection and analysis of relevant data either directly or by support, and these include the

OECD, the World Bank, the European Commission. The next section considers the GHGs particularly human activity-related CO₂, and where in the world they come from.

V Global Greenhouse Gas Emissions

The discussions of GHG emissions in terms of climate change often concentrates on CO₂ although this is just one of the main GHGs along with water vapour (H₂O), methane (CH₄), nitrous oxide (N₂O) and fluorinated gases (F-gases) that is the collective term for hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulphur hexafluoride (SF₆), and nitrogen trifluoride (NF₃), and the composition of GHG emissions is very different in various countries – but CO₂ is recognised as the largest contributor to global warming (National Atmospheric Emissions Inventory, 2025). The variety of proportions of different GHGs emitted by the major CO₂ emitting countries (the top 20 emitters of CO₂ ranked in order of those CO₂ emissions) is clear in the table below (Table 1) that summarises data compiled by the Joint Research Centre (JRC), the European Commission’s science and knowledge service, using the Emissions Database for Global Atmospheric Research (EDGAR) in collaboration with the International Energy Agency (IEA) (European Commission, 2024). The integrity of data is of crucial importance in the analysis and there have been efforts to overcome some of the challenges of data collection especially in non-OECD countries by collaborating with national authorities to improve data quality (European Commission, 2024). EDGAR provides GHG emissions time series for all countries and for all anthropogenic sectors from 1970 until 2023, including emissions and removals from land and forestry, and the JRC states that their 2024 Policy Report, ‘GHG Emissions of All World Countries: 2024’ contributes to the Paris Agreement process with an independent quantitative overview of global GHG emissions.

Table 1

CO₂ and GHG Emissions by Country in 2023 (the 20 Biggest Emitters)												
Sources: European Commission (2024) and EC/JRC/IEA (2024)												
Position	Nation	GHG Emissions Mn Metric Tons	GHG Proportions				GHG Emissions by gas type				Total	% of World CO ₂ in 2023
			CO ₂	N ₂ O	CH ₄	F-gases	CO ₂	N ₂ O	CH ₄	F-gases		
1	China	15,943.99	83.2%	2.6%	11.2%	3.0%	13265.40	414.54	1785.73	478.32	15943.99	35.84%
2	United States	5,960.80	78.6%	3.5%	14.3%	3.6%	4685.19	208.63	852.39	214.59	5960.80	12.66%
3	India	4,133.55	71.5%	6.5%	20.3%	1.7%	2955.49	268.68	839.11	70.27	4133.55	7.99%
4	Russia	2,672.04	77.5%	2.8%	17.5%	2.2%	2070.83	74.82	467.61	58.78	2672.04	5.60%
5	Japan	1,041.01	90.8%	1.5%	5.5%	2.2%	945.24	15.62	57.26	22.90	1041.01	2.55%
6	Iran	996.75	78.1%	3.0%	17.0%	1.9%	778.46	29.90	169.45	18.94	996.75	2.10%
7	Indonesia	1,200.20	56.2%	6.6%	35.8%	1.4%	674.51	79.21	429.67	16.80	1200.20	1.82%
8	Saudi Arabia	805.16	77.4%	1.3%	14.0%	7.3%	623.19	10.47	112.72	58.78	805.16	1.68%
9	Germany	681.81	85.5%	4.4%	8.7%	1.4%	582.95	30.00	59.32	9.55	681.81	1.58%
10	Canada	747.68	76.9%	4.5%	16.4%	2.2%	574.96	33.65	122.62	16.45	747.68	1.55%
11	South Korea	653.85	87.7%	2.2%	6.8%	3.3%	573.42	14.38	44.46	21.58	653.85	1.55%
12	Mexico	712.10	68.4%	5.3%	23.2%	3.1%	487.08	37.74	165.21	22.08	712.10	1.32%
13	Brazil	1,300.17	36.9%	12.7%	49.1%	1.3%	479.76	165.12	638.38	16.90	1300.17	1.30%
14	Turkey	606.43	72.3%	5.6%	20.7%	1.4%	438.45	33.96	125.53	8.49	606.43	1.18%
15	South Africa	522.12	76.1%	3.8%	17.0%	3.1%	397.33	19.84	88.76	16.19	522.12	1.07%
16	Australia	571.84	65.3%	8.3%	24.2%	2.2%	373.41	47.46	138.39	12.58	571.84	1.01%
17	Vietnam	524.13	71.2%	4.9%	22.0%	1.9%	373.18	25.68	115.31	9.96	524.13	1.01%
18	Italy	374.12	81.7%	3.8%	10.5%	4.0%	305.66	14.22	39.28	14.96	374.12	0.83%
19	United Kingdom	379.32	79.6%	5.5%	12.1%	2.8%	301.94	20.86	45.90	10.62	379.32	0.82%
20	Poland	363.79	78.9%	6.6%	13.1%	1.4%	287.03	24.01	47.66	5.09	363.79	0.78%

An examination of human activity-related CO₂ emissions at national level in 2023 (see Table 2 below) shows that there was quite a clear concentration among certain countries with China and the United States responsible for by far the largest proportions of the world's CO₂ emissions, with just under 36% for China and 12.66% for the United States. It is also significant that a very small number of nations, just the top 5 CO₂ emitting countries were the source of 64.6% of the world's total human-induced CO₂ emissions and the top 10 countries accounted for 73.4% of the world's human activity related CO₂ emissions as may be seen in Table 2 below.

Further analysis of the figures from 2023 (European Commission, 2024) indicates that just 20 nations were responsible for over 84% of the world's human activity related CO₂ emissions - again this is shown in Table 2 below – and the nations are ordered according to the CO₂ emissions not by overall GHG emissions, and this is because the greatest concern is over CO₂ that has the largest global warming impact (National Atmospheric Emissions Inventory, 2025).

Table 2

CO₂ Emissions by Country in 2023				
Sources: European Commission (2024) and EC/JRC/IEA (2024)				
Position	Nation	CO₂ Emissions	Percentage of	
	(20 Biggest Emitters)	Million Metric Tons	World Emissions*	
1	China	13,265.40	35.84%	
2	United States	4,685.19	12.66%	
3	India	2,955.49	7.99%	
4	Russia	2,070.83	5.60%	
5	Japan	945.24	2.55%	64.64%
6	Iran	778.46	2.10%	
7	Indonesia	674.51	1.82%	
8	Saudi Arabia	623.19	1.68%	
9	Germany	582.95	1.58%	
10	Canada	574.96	1.55%	73.38%
11	South Korea	573.42	1.55%	
12	Mexico	487.08	1.32%	
13	Brazil	479.76	1.30%	
14	Turkey	438.45	1.18%	
15	South Africa	397.33	1.07%	
16	Australia	373.41	1.01%	
17	Vietnam	373.18	1.01%	
18	Italy	305.66	0.83%	
19	United Kingdom	301.94	0.82%	
20	Poland	287.03	0.78%	84.23%

VI Carbon Pricing Systems around the World

There are broadly three main forms of carbon pricing: carbon taxes, emissions trading systems and carbon crediting mechanisms. The theoretical bases that often inform government policies will be discussed in brief terms before moving on to a discussion about where they are being applied in global carbon markets (for a more detailed discussion see Kirkpatrick, 2022).

Carbon Taxes – A Brief Outline and Basic Theory

Carbon Taxes are indirect taxes as they are usually paid to governments by producers and are likely to be borne by the end consumer who is therefore indirectly paying the carbon tax. Carbon taxes are also likely to have an inflationary effect, at least a temporary inflationary effect that may become permanent depending for example on how a government uses the tax revenues. Carbon taxes may be considered as the creation of a price signal in response to negative externalities (Pigou, 1920 and later contributors) although the problems of tax distortions and deadweight loss have been introduced and analysed by for example, Hicks (1941) and Harberger (1964 and 1971) while the relevance and efficacy of carbon taxes has also been challenged (Coase, 1960). Carbon taxes may also be unpopular if the taxpayers perceive that such forms of carbon pricing threaten national competitiveness and the economic growth objective. The problem of ‘free-riding’ by nations that do not impose their own carbon taxes has been discussed by for example, MacKay et al. (2017) and Nordhaus (2015 and 2017) who suggested that such concerns were a driver of the creation of ‘climate clubs’. Carbon Border Adjustment Mechanisms (CBAMs), discussed in section VIII below) are a response to concerns about ‘free-riding’.

Emissions Trading Systems or ETSs – A Brief Outline and Basic Theory

ETSs are otherwise known as ‘cap-and-trade’ systems because they work by a regulatory body such as a government department setting a cap on carbon emissions or other GHG emissions and distributing allowances to businesses that may then trade those allowances that are often in the form of certificates. ETSs and carbon taxes both serve to ‘put a price’ on CO₂ emissions but there are key differences in that while carbon taxes may prioritise certainty of price, ETSs prioritise certainty of quantity so the market determines the price based on forces of demand and supply. It is important to point out that in practice these systems may not conform exactly to the theoretical expectations and there are also variants of the cap-and-trade instrument where there may be a hybrid approach so that an ETS may have features of price control such as a guaranteed ceiling and/or floor price (Kirkpatrick, 2022).

Carbon Crediting Systems – A Brief Outline and Basic Theory

Carbon Crediting Systems share more features with ETSs than with carbon taxes. Like ETSs Carbon Crediting Systems are designed to have an impact on the quantity of CO₂ emissions rather than impacting price directly like carbon taxes. Carbon credits are created by projects for various GHGs including reducing or removing CO₂ from the atmosphere, including sustainable agriculture, peatland restoration, reforestation and afforestation, and carbon capture and storage. Carbon crediting is discussed more in Section VII below.

Table 3 below, provides a summary of major CP systems as at 2025.

Table 3

Carbon Pricing Instruments around the World (2025)			
Region / Countries	Instrument Type (Year of Implementation)	Source: Compiled from sources in World Bank (2025b)	
	Emissions Trading System	Carbon Tax	Crediting Mechanism
The Americas			
Canada	Canada Federal OBPS (2019) Newfoundland & Labrador PSS (2019) Nova Scotia OBPS (2019) Ontario EPS (2022) Quebec CaT (2013) Saskatchewan (2019) New Brunswick (2021) Alberta TIER (2007) British Columbia OBPS (2016)	Northwest Territories (scope reduced 2025)	Canada Federal GHG Offset System (2022) Quebec Offset Crediting Mechanism (2013) Alberta Emission Offset System (2007) British Columbia Offset Program (2016)
USA	RGGI (2009) Colorado GHG Crediting Trading System (2025) Washington (State) CCA (2023) Oregon ETS (2025) California CaT (2012)		RGGI CO2 Offset Mechanism (2005) Washington (State) Crediting Mechanism (2023) California Compliance Offset Program (2013)
Mexico	Mexico ETS (2020)	Mexico Federal Carbon Tax (2014) State of Mexico Carbon Tax (2024) Durango Carbon Tax (2024) Zacatecas Carbon Tax (2017) Guanajuato Carbon Tax (2023) Tamaulipas Carbon Tax (2022) Colima Carbon Tax (2025) Yucetan Carbon Tax (2022) San Luis Potosi Carbon Tax (2025)	
Colombia		Colombia Carbon Tax (2017)	Colombia Crediting Mechanism (2017)
Chile		Chile Carbon Tax (2017)	Chile Green Tax Emissions Offsetting Scheme (2024)
Argentina		Argentina Carbon Tax (2018)	
Europe			
UK	UK ETS (2021)	UK Carbon Price Support (2013)	Peatland Code (2015) UK Woodland Carbon Code (2011)
Norway	EU ETS (2005) even though in EEA not in EU	Norway Carbon Tax (1991)	
Iceland	EU ETS (2005) even though in EEA not in EU	Iceland Carbon Tax (2010)	
Switzerland	Switzerland ETS (2008)	Switzerland Carbon Tax (2008)	CO2 Attestations Crediting Mechanism (2012)
EU (See Note Below)	EU ETS (2005)		
NOTE: A number of EU member states also have their own Carbon Pricing Instruments that operate alongside the EU ETS in those states			
Ukraine		Ukraine Carbon Tax (2011)	
Middle East			
Saudi Arabia			Greenhouse Gas Crediting and Offsetting Mechanism (2024)
Africa			
South Africa		South Africa Carbon Tax (2019)	South Africa Crediting Mechanism (2019)
Asia			
China	China National ETS (2021) Chongqing Pilot (2014) Hubei Pilot ETS (2014) Tianjin Pilot ETS (2013) Shanghai Pilot ETS (2013) Fujian Pilot ETS (2016) Shenzhen Pilot ETS (2013)		China GHG Voluntary Emission Reduction Program (2012) Chongqing Carbon Offset Mechanism (2021) Fujian Forestry Offset Crediting Mechanism (2016)
Taiwan, China		Taiwan Carbon Fee (2024)	Taiwan Voluntary GHG Reduction Program (2015)
Japan		Japan Carbon Tax (2012)	Japan Joint Crediting Mechanism (2013) J-Credit Scheme (2013) Tokyo Cap-and-Trade Program (2010)
	Tokyo CaT (2010)		
Korea Rep.	Korea ETS (2015)		Republic of Korea Offset Credit Mechanism (2015)
Thailand			Thailand Voluntary Emission Reduction program (2014)
Indonesia	Indonesia ETS (2023)		
Kazakhstan	Kazakhstan ETS (2013)		Kazakhstan Crediting Mechanism (2013)
Sakhalin Oblast (Russia)			Sakhalin Oblast Pilot Crediting Mechanism (2022)
Australasia			
Australia	Australia Safeguard Mechanism (2023)		Australia Carbon Credit Unit Scheme (ACCU) (2012)
New Zealand	New Zealand ETS (2008)		

Emissions Trading

The Kyoto Protocol (1997) refers to the development of emissions trading including rules and guidelines to be defined in the Conference of the Parties. Emissions trading is one of the three Kyoto Mechanisms discussed earlier and as discussed, is one of the major forms of carbon pricing.

ETs have been established at various levels of government including cities, through provinces and states to countries and then a supranational level in the form of the EU ETS (see Table 4, below).

Table 4

Emissions Trading Systems - At Different Levels of Government						
Cities	Provinces & States				Countries	Supranational
	Canada	China	Japan	United States		
Beijing	Alberta	Fujian	Prefecture Saitama	California	Australia	European Union
Chongqing	British Columbia	Guangdong	Tokyo	Connecticut	Austria	
Shanghai	New Brunswick	Hubei		Delaware	Canada	
Shenzhen	Newfoundland			Maine	China	
Tianjin	and Labrador			Maryland	Germany	
	Nova Scotia			Massachusetts	Indonesia	
	Ontario			New Hampshire	Kazakhstan	
	Quebec			New Jersey	Mexico	
	Saskatchewan			New York	Montenegro	
				Rhode Island	New Zealand	
				Vermont	Rep of Korea	
				Washington	Switzerland	
					United Kingdom	

Source: ICAP (2024a) 'Emissions Trading Worldwide: Status Report 2024'

Some of the major emissions trading systems (ETs) are discussed briefly below – note that there is more discussion of these systems in the context of the global carbon markets in Section VII below.

The China National ETS was introduced in 2021 with the objective of contributing to the effective control and gradual reduction of CO₂ emissions with an annual cap that has risen from around 4,500 MtCO₂ in 2019 and 2020, to round 5,100 MtCO₂ in 2021 and 2022 and then around 5,200 MtCO₂ in 2023 (ICAP, 2024b).

USA – California Cap-and-Trade Program began operation in 2012 and covers around 76% of California State's GHG emissions with a cap of 267.4 MtCO₂e or MtCO₂ in 2025 (ICAP, 2024b).

USA – Regional Greenhouse Gas Initiative launched in 2009 as the first mandatory GHG ETS in the United States involving 10 states: Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Rhode Island, and Vermont, and in 2020 covering around 14% of aggregate states' emissions with a cap of 63 MtCO₂ in 2024 (ICAP, 2024b). There are three other, separate ETs in the USA (discussed in Section VII).

The EU Emissions Trading System (EU ETS) that was introduced in 2005 was hailed as the world's first of its type and remained one of the largest twenty years later in 2025. The EU ETS that currently has 30 members (see Table 5 below) – the 27 EU members plus 3 European Economic Area (EEA) members - came into being through the adoption of the EU ETS Directive of 2003.

Table 5

EU Emission Trading System (EU ETS) Membership in 2024		
State	EU/EEA/Other	Comments
Austria	EU	
Belgium	EU	
Bulgaria	EU	
Croatia	EU	
Cyprus (Republic of)	EU	
Czech Republic	EU	
Denmark	EU	
Estonia	EU	
Finland	EU	
France	EU	
Germany	EU	
Greece	EU	
Hungary	EU	
Iceland	EEA	Also allowed to be in EU single market
Ireland (Republic of)	EU	
Italy	EU	
Latvia	EU	
Liechtenstein	EEA	Also allowed to be in EU single market
Lithuania	EU	
Luxembourg	EU	
Malta	EU	
Netherlands	EU	
Northern Ireland (UK)	UK (see Note 1)	Northern Ireland Protocol (see below)
Norway	EEA	Also allowed to be in EU single market
Poland	EU	
Portugal	EU	
Romania	EU	
Slovakia	EU	
Slovenia	EU	
Spain	EU	
Sweden	EU	
Switzerland	EFTA (See Note 2) (Not in EU or EEA)	But has bilateral agreements with EU to allow it to partial access to the EU single market.
Note 1 Under the Northern Ireland Protocol (1921) and the Windsor Framework (2023) aiming to provide safeguards to the EU Single Market to which Northern Ireland (in UK) has unique access, Northern Ireland electricity generators continue as part of the EU ETS Note 2 The Swiss ETS has been linked to the EU ETS since January 2020 and the two systems adhere to similar rules ensuring compatibility. Sources: UK Government at www.gov.uk and European Parliament at www.europarl.europa.eu		

Northern Ireland electricity generators also continue to be part of the EU ETS despite Brexit under the special terms of the Northern Ireland Protocol (2021) and the Windsor Framework (2023). Switzerland is neither an EU member or a EEA member but it is a European Free Trade Association (EFTA) member and with bilateral agreements its bilateral agreements with the EU mean that it has partial access to the EU single market, furthermore the Swiss ETS is linked to the EU ETS as it uses the same benchmarks for entities covered by the Swiss ETS. The EU ETS cap was reduced during the early 2020s from 1,571.6 MtCO₂ in 2021 to 1,386.1 MtCO₂ in 2024 (ICAP, 2024b). A number of EU member states that are members of the EU ETS also operate their own national CP instruments as summarised in Table 6 below.

Table 6

EU Member States in EU ETS also with own Carbon Pricing Instruments			
	ETS	Carbon Tax	Crediting Mechanism
Sweden	EU ETS (2005)	Sweden Carbon Tax (1991)	
Finland	EU ETS (2005)	Finland Carbon Tax (1990)	
Estonia	EU ETS (2005)	Estonia Carbon Tax (2000)	
Latvia	EU ETS (2005)	Latvia Carbon Tax (2004)	
Poland	EU ETS (2005)	Poland Carbon Tax (1990)	
Germany	Germany ETS (2021) EU ETS (2005)		
France	EU ETS (2005)	France Carbon Tax (2014)	
Spain	EU ETS (2005)	Spain Carbon Tax (2014)	Spain FES-CO ₂ Program (2011)
Austria	Austria ETS (2022) EU ETS (2005)		
Hungary	EU ETS (2005)	Hungary Carbon Tax (2023)	
Portugal	EU ETS (2005)	Portugal Carbon Tax (2015)	Portuguese Voluntary Carbon Market (2024)
<i>Source: World Bank (2025b) Carbon Pricing Dashboard - https://carbonpricingdashboard.worldbank.org</i>			

The UK Emissions Trading Scheme (UK ETS) that came into operation on 1 January 2021, has its origins in the EU ETS that was introduced in 2005. The UK was a founding member of the EU ETS.

VII Global Carbon Pricing Impact and Markets

Carbon Pricing Impact and Coverage

The World Bank reported in 2025 that the share of global emissions covered by direct CP mechanisms in the form of carbon taxes, carbon crediting and ETSs, had grown to around 28% with 80 carbon taxes or ETSs in operation that could be at a national or sub-national – city or province - level of operation (World Bank, 2025a). The figure of 80 direct CP mechanisms seems relatively small but as the World Bank points out, indirect carbon pricing policies such as fuel excise taxes, fossil fuel subsidies, and differentiated VAT rates can also provide strong price signals to change the economics of high-emissions fuels or products (World Bank, 2025a). **Table 7** below shows major CP systems and the emissions covered. The World Bank has also reported how the energy crisis starting in 2022 has led to delays in the introduction or expansion of carbon taxes or ETSs in Indonesia, Austria and Germany, and to Slovenia repealing its carbon tax that had been in place since 1996, while the energy

Table 7**Carbon Pricing Systems and Emissions Covered**

Sources: World Bank and European Commission in association with IEA (see Note at bottom of the table)

Nation	Share (%) of World CO2 Emissions*	State/sub national or provincial jurisdiction	Carbon Tax Coverage (percentage)			Emissions Trading System Coverage (percentage)			Combined Global Share (Estimate)
			Nation	State	World	Nation	State	World	
China	35.84%	China National ETS				51.00%		15.29%	
		Shenzen					37.00%	0.04%	
		Shanghai					21.00%	0.10%	
		Beijing					17.00%	0.05%	
		Guangdong					12.00%	0.18%	
		Tianjin					18.00%	0.07%	
		Hubei					20.00%	0.14%	
		Chongqing					14.00%	0.06%	
		Fujian					16.00%	0.10%	16.02%
United States	12.66%	California CaT					76.00%	0.56%	
		Washington CCA					71.00%	0.12%	
		Oregon ETS					48.00%	0.05%	
		RGGI					14.00%	0.17%	
		Colorado ETS					3.00%	0.00%	0.89%
European Union	7.20%	EU ETS (multi-state)				40.00%		2.48%	2.48%
Japan	2.55%	Japan Carbon Tax	80.00%		1.57%				
		Tokyo CaT					18.00%	0.02%	
		Saitama ETS					18.00%	0.01%	1.61%
Indonesia	1.82%	Indonesia ETS				24.00%		0.54%	0.54%
Germany	1.58%	Germany ETS Also in EU ETS				39.00%		0.50%	0.50%
South Korea	1.55%	Korea ETS				79.00%		0.98%	0.98%
Canada	1.55%	Canada Fed OBPS				3.00%		0.04%	
		Alberta TIER					59.00%	0.29%	
		BC OBPS					26.00%	0.03%	
		Quebec CaT					76.00%	0.11%	
		Newfoundland & Lab					36.00%	0.01%	
		Northwest Territories		76.00%	0.00%				
		Nova Scotia OBPS					36.00%	0.01%	
		Saskatchewan OBPS					43.00%	0.05%	
		New Brunswick OBPS					54.00%	0.01%	
		Ontario ETS					26.00%	0.07%	0.60%
Mexico	1.32%	Mexico Carbon Tax	13.00%		0.17%				
		Mexico ETS				36.00%		0.48%	
		Zacatecas Carbon Tax		50.00%	0.01%				
		Queretaro C Tax		18.00%	0.01%				
		State of Mexico Ctax		47.00%	0.10%				
		Yucatan Carbon Tax		32.00%	0.01%				
		Guanajuato C Tax		42.00%	0.03%				0.80%
South Africa	1.07%	South Africa C Tax	82.00%		0.81%				0.81%
Australia	1.01%	Safeguard Mechanism				26.00%		0.28%	0.28%
United Kingdom	0.82%	Carbon Price Support	13.00%		0.10%				
		UK ETS				27.00%		0.19%	0.30%
Poland	0.78%	Poland Carbon Tax	24.00%		0.16%				
		Also in EU ETS							0.16%

Note:

Data is drawn from or based on information produced by the World Bank 'Carbon Pricing Dashboard' in 2025 analysing data available in 2025 and pointing out that prices are not directly comparable due to differences in coverage, compliance and compensation arrangements. Data on 'Share of World CO2 Emissions' uses data for 2023 reported by European Commission (2024)

Important note - all figures are based on the external sources indicated and may be subject to later revisions.

crisis also seems to have played a part in the delays and a court challenge following the US State of Pennsylvania's passage of legislation to join the Regional Greenhouse Gas Initiative (RGGI) (World Bank, 2023).

Carbon Markets

The development of carbon pricing mechanisms in the form of emissions trading systems has allowed carbon markets to develop. There are broadly two forms of carbon markets: the first, 'mandatory carbon markets' or 'compliance carbon markets (CCMs)' that include the China National ETS, the EU ETS and the UK ETS that have been created and subsequently are regulated by nations or states (as in the USA) or in the case of the EU, a union of national authorities; and the second, 'voluntary carbon markets' (VCMs) are not regulated.

Mandatory or Compliance Carbon Markets (CCMs)

For CCMs the commodity is in the form of carbon allowances or carbon credits. Carbon allowances are permits to emit CO₂ and as facilitated by a cap-and-trade system. Carbon credits are also created by governments, at least that is usually the case, and they are a measurement unit by which a business or organisation may be repaid for investing in carbon-reducing initiatives or 'green projects'. CCMs are regulated in accordance with a specific ETS to comply with the country's NDCs under the Paris Agreement (2015) with mandatory participation by 'in scope' entities.

Carbon prices in CCMs are determined by processes of the specific ETS operating according to its emissions cap, and the market forces of supply and demand for emissions allowances in international carbon markets. The participants are the businesses that are 'in scope' and issues with carbon allowances or credits within the ETS. The carbon allowances usually take the form of certificates each granting permission to generate one ton of CO₂ emissions. The marketplace is a regulated market set by cap-and-trade regulations at regional or state level. The exchanges such as the Intercontinental Exchange (ICE) and the European Energy Exchange (EEX) offer platforms for continuous trading so carbon allowances can be traded throughout the day.

The mandatory carbon markets in some of the major CO₂ emitting jurisdictions are discussed briefly below – please note that at the time of writing many countries are considering the introduction of CP instrument and it is not practical to consider all nascent markets here. It is worth pointing out that China, USA and the EU nations together account for significantly more than half of the world's total anthropogenic CO₂ emissions (World Bank, 2025b).

China: China National Emissions Trading Scheme (or China National Carbon Market) started operating in 2021. It was reported in 2024 to be the largest emissions trading system operating in the world in terms of covered emissions that are estimated to be around 8 billion tCO₂ (ICAP, 2024b). The China National ETS covers 51 per cent. of the jurisdiction's emissions based on data available by 2025 (World Bank, 2025b). There are also Pilot ETSs operating in a number of the Chinese provinces that are as follows (with coverage of jurisdiction CO₂ emissions in brackets: Beijing (17%); Chongqing (14%); Fujian (16%); Guangdong (12%); Hubei (20%); Shanghai (21%); Shenzhen (37%); and Tianjin (18%).

USA: In the USA there are a number of ETSs operating in different states but there is to date no overarching nationwide ETS.

The California Cap-and-Trade or 'CaT' programme is a market-based system with a declining limit on CO₂ emissions set by the California Air Resources Board (CARB). The California Global Warming Solutions Act (USA) started operating in 2012 with a jurisdiction that covers only the State of California. Sectors regulated in the First Phase were large industrial facilities such as cement, glass, hydrogen, iron and steel; electricity generation; electricity imports; other stationary combustion; CO₂ suppliers; and then in a Second Phase further sectors were added: suppliers of natural gas; suppliers of certain distillate fuel oils; suppliers of liquid petroleum gas; and, suppliers of liquefied natural gas.

The Washington Climate Commitment Act signed into law in 2021 led to Washington State's ETS or 'cap-and-invest program' starting to operate in January 2023 covering 71% of the state's emissions (see Table 7) with a cap trajectory to reduce the state's GHG emissions to a level 95% below 1990 levels by 2050. Sectors covered are mining and extractives, transport, buildings, industry and power (ICAP, 2024b).

The Oregon ETS started operating in 2025 with a cap for GHGs of 24.1 MtCO₂e in 2025 and covering the sectors: agriculture and forestry, maritime, mining and extractives, waste, transport, buildings and industry (ICAP, 2024b).

The Regional Greenhouse Gas Initiative (RGGI) was formed in 2009 and was the first mandatory carbon market GHG ETS in the USA and initially covered ten states and though there have been some changes over the years – for example Virginia joined in 2021 but left in 2023 - the founding states are still the membership in 2025: Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Rhode Island, and Vermont. The RGGI had a cap of 63 MtCO₂ in 2024 covering the power sector (ICAP, 2024b).

The Colorado ETS was formed when the Colorado Air Quality Control Commission or AQCC adopted regulations to establish an ETS for large manufacturers in October 2023 and this scope is to be expanded following a decision in December 2024 to include oil and gas operations from 2028. A distinguishing feature of the Colorado ETS is that it uses a hybrid approach combining both absolute emissions and emissions-intensity targets. The ETS is based on regulations for Greenhouse Gas Emissions and Energy Management for Manufacturing (GEMM) that were developed after 2021 so that GEMM 1 a category covering energy-intensive and trade-exposed manufacturers emitting 25,000 tCO₂e or more annually and is regulated based on emissions intensity with targets for improving GHG per unit of output. GEMM 2 is a second category covering all other manufacturing facilities emitting 25,000 tCO₂e or more annually, regulated by reference to absolute emissions and subject to an emissions cap.

EU/Europe: The European Market for Carbon Credits was the world's first and by 2025 among the largest carbon markets in the world (European Commission, 2025). The EU ETS provides carbon credit contracts for trading and delivering European Union Allowances (EUAs) where one EUA allows the holder to emit one ton of CO₂ or CO₂ equivalent in greenhouse gas. The EUA Market has both a primary market and a secondary market. The primary market includes the free allocation of a portion of the total EUAs to operators, a specific fund reserve for certain projects and all remaining EUAs are sold through auctions that are run by auction platforms for continuous trading mostly on the Intercontinental Exchange (ICE) and the European Energy Exchange (EEX). EUAs are classed as derivatives

and can be traded in spot, forward and future markets. MiFID regulations need to be observed.

Japan: With its significant economy and strategic position in Asia, Japan seems to be poised to occupy a pivotal position in carbon markets in the region. In 2025 there are two major ETSs in operation in Japan: one in Tokyo and the other in Saitama.

The Tokyo Cap-and-Trade or 'Tokyo CaT' program was launched in April 2010 as Japan's first mandatory ETS covering around 20% of the metropolitan area's CO₂ emissions from large buildings, factories, heat suppliers and other facilities consuming large quantities of fossil fuels. The 2024 Cap was 12.2 MtCO₂e and there are targets to achieve a 50% reduction from 2000 GHG levels by 2030, and Net Zero CO₂ by 2050.

The Saitama Prefecture's ETS was launched in April 2011 and covers 16% of the carbon emissions from buildings and industry in the jurisdiction. There is a GHG target to achieve a 46% reduction from 2013 Financial Year levels by 2030.

Indonesia: As a large nation with a significant economic position in Asia it is not surprising that Indonesia is within the top ten CO₂ emitters based on data in 2023 presented and analysed by IEA and the EC researchers in JRC (European Commission, 2024) – it is sixth in the world in terms of GHG emissions (see Table 1 above) and seventh in the world in terms of CO₂ emissions (see Table 2 above). Indonesia launched an ETS in 2023 described as 'Economic Value of Carbon' or 'Nilai Ekonomi Karbon' (NEK) specifically for coal fired power plants connected to the Perusahaan Listrik Negara (PLN) grid and the system was reported to cover 99 coal-fired power plants accounting for 81.4% of the nation's power generation capacity (ICAP, 2023a). In 2024 the ETS scope was expanded to cover installations with a capacity of 25 MW or more, so that added a further 47 coal-fired power plants to the ETS (ICAP, 2024b). The Indonesia NEK ETS covers three GHGs: CO₂, CH₄ and N₂O. In Phase 1 (2023 and 2024) the total emissions limit was approximately 256.8 MtCO₂e. The emissions limits for Phase 2 (2025 – 2027) and Phase 3 (2028 to 2030) have not yet been determined.

South Korea: The Korea Emissions Trading System (K-ETS) started operating in 2015 covering several GHGs: CO₂, CH₄, N₂O, HFCs, PFCs, and SF₆ (ICAP, 2024b). South Korea (or the Republic of Korea) is one of the nations that has a significant reliance on carbon-intensive manufacturing and it ranks sixth in the world in terms of GHG emissions (see Table 1 above) and seventh in the world in terms of CO₂ emissions (see Table 2 above). The Korea ETS covers a wide range of sectors: maritime, waste, domestic aviation, transport, buildings, industry and power. Phase 1 (2015-2017) and Phase 2 2018-2020) have been completed so the ETS is now in Phase 3 (2021-2025) with a cap of 3,048.3 MtCO₂e corresponding to an average annual cap of 610 MtCO₂e including reserves (ICAP, 2024b).

Canada: The oil and gas sector, transportation, heavy industry, buildings, agriculture, electricity generation, and waste are the main contributors to carbon emissions in Canada (Government of Canada, 2025) and the scale and intensity of these activities put Canada in tenth place globally in terms of GHG emissions and CO₂ emissions (see Table 1 and Table 2 above). The oil and gas sector is Canada's largest source of GHG emissions and on 7 December 2023, the Canadian Government announced the development of a federal cap-and-trade system for oil and gas emissions with an emissions cap in the range 106-112 MtCO₂e (to be finalised) to be phased in between 2026 and 2030, consistent with the

overall plan to achieve ‘net zero’ by 2050 (ICAP, 2023b). Canada does not as of 2025 have a single national ETS as such but operates a dual pricing system with a federal backstop in the Canada Federal Output-Based pricing System (OBPS) that allows the provinces and territories to implement their own systems subject to agreed national standards (ICAP, 2024b). As at 2025 the Canada Federal OBPS applied in the provinces of Manitoba and Prince Edward Island, and in the territories of Nunavut and Yukon. As at 2025 in Canada there are separate ETSs in the eight Canadian provinces other than Manitoba and Prince Edward Island, these are: Alberta, British Columbia, New Brunswick, Newfoundland and Labrador, Nova Scotia, Ontario, Quebec, and Saskatchewan. Northwest Territories is the one jurisdiction in Canada operating a carbon tax – there is no ETS in the territory – and the Government of Northwest Territories announced in March 2025 that it would be removing the carbon tax for all consumers except large emitters, specifically the diamond mines confirmed as the territory’s only large emitters, on 1 April 2025 (Government of the Northwest territories, 2025).

A Consideration of Outcomes of CCMs in Terms of Prices and Coverage

Two important measures in the analysis of a carbon pricing system are the price level and the coverage in terms of the proportion of the jurisdiction’s carbon emissions. There is a considerable range in both pricing and coverage of the CP instruments in the biggest CO₂-emitting jurisdictions in 2025 as shown in Table 8 below.

Table 8

Carbon Pricing and Emissions Coverage by Jurisdiction (concentrating on biggest CO ₂ emitters)						
Jurisdiction and Descriptor	Instrument Category	Carbon Tax Price (USD)	ETS Price (USD)	Coverage (%) Jurisdiction	Coverage (%) Global	Comments Prices are in US\$/tCO ₂ e
China National ETS	ETS		11.76	51%	15.29%	China also has city/provincial ETSs
EU ETS	ETS		70.37	40%	2.48%	Some states also have own pricing
USA - Washington CCA	ETS		50.00	71%	0.12%	USA has no national ETSs
USA - California CaT	ETS		29.27	76%	0.56%	Just some state level ETSs
USA - RGGI	ETS		23.32	14%	0.17%	There are 11 states in the RGGI
USA - Oregon CCP	ETS		N/A	48%	0.05%	State Law under consideration
USA - Colorado GHG CTS	ETS		N/A	3%	0.00%	
Japan Carbon Tax	Carbon Tax	1.93		80%	1.57%	
Japan - Tokyo CaT	ETS		4.01	18%	0.02%	
Japan - Saitama ETS	ETS		0.96	18%	0.01%	
Indonesia ETS	ETS		0.72	24%	0.54%	
Germany - National ETS	ETS		48.55	39%	0.50%	Operates for transport & buildings
Canada Federal OBPS	ETS		66.21	3%	0.04%	Canada also has provincial ETSs
South Korea ETS	ETS		6.45	79%	0.98%	
Mexico Carbon Tax	Carbon Tax	3.94		13%	0.17%	Also individual state Carbon Taxes
Mexico ETS	ETS		0.00	36%	0.48%	Zero CP - ETS is still in Pilot Phase
South Africa Carbon Tax	Carbon Tax	12.85		82%	0.81%	
Australia Safeguard Mech ETS	ETS		21.82	26%	0.28%	
UK Carbon Price Support	Carbon Tax	23.24		13%	0.10%	
UK- ETS	ETS		57.23	27%	0.19%	
Poland Carbon Tax	Carbon Tax	0.00		24%	0.16%	Zero - as it interacts with EU ETS

As at 2025 there are a number of jurisdictions considering carbon pricing instruments - the above list is indicative and not exhaustive
Source: World Bank Dashboard 2025 - Note prices are as at 1 April 2025 or are the latest available prior to 1 April 2025.

The data shown in Table 8 uses the World Bank’s Carbon Pricing Dashboard (World Bank, 2025b) to obtain information about carbon price levels and coverage across the world’s

jurisdictions and that provides an opportunity to compare the different jurisdictions although some caution is necessary in interpreting the data as there may be variations in the completeness of data collection, specific methodologies for data collection, and the pricing mechanisms even within the broader categories of ETSs and carbon taxes. The data is drawn from CCMs and although VCMs (discussed in the next section) may be part of the carbon pricing story it is much harder to obtain reliable and consistent or informative data as VCMs lack standardised reporting, necessary transparency and consistent methodologies. The next section considers VCMs and includes a comparison of the features of CCMs and VCMs that may help in further analysis of their contributions in advancing carbon pricing policies.

Voluntary Carbon Markets VCMs)

Certain types of products are relevant for investment in the VCMs. It has been argued that nature-based carbon markets can provide a significant contribution to 'carbon mitigation' required by 2030 to keep the 1.5 degree target within reach (Moore et al, 2023). Although there is no formal or direct regulation of VCMs there have been some discussions about introducing measures to ensure there is transparency and integrity in the VCMs are they do not become vehicles for greenwashing.

In VCMs the commodity is the form of 'carbon offsets' resulting from carbon sequestration to remove CO₂ from the Earth's atmosphere and hold it in solid or liquid form. The VCM is self-regulated so there is no formal or direct regulation by governments or financial authorities. The VCM is a nascent market and there are issues to consider, for example, there are discussions in most jurisdictions about how matters such as transparency and integrity need to be addressed. Carbon prices in VCMs are determined in a specific project with chosen participants. The prices are driven by supply and demand as well as specific project characteristics, buyer preferences and the quality of each project, so for example, those projects certified under rigorous standards for biodiversity, community contribution, and climate impact are more likely to command higher prices.

The participants in VCMs are businesses that take the opportunity to work with similarly motivated businesses or individuals to offset their carbon emissions. Throughout the 2020s here has been increasing interest in the VCM by participants including oil and gas majors, hedge funds and banks especially those that have financial structuring and commodity trading expertise. The VCMs may be distinguished from the CCMs that have organised regulated markets, as VCMs do not have a centralized market. In a VCM project developers or businesses sell carbon offsets to buyers through brokers, retailers or using digital platforms.

A significant development in VCMs has come in the form of initiatives by Shell and Boston Consulting Group that show the strategic importance of major purchasers of carbon offsets in the VCM. It has been estimated that the VCM is growing and is set to grow more although precise estimates are hard to pin down, for example the Boston Consulting Group estimated that the value of the VCM could reach between USD10 billion and USD40 billion by 2030 (Porsberg-Smith et al, 2023). Carbon pricing can seem like a largely national decision but there is scope for most nations to affect and be affected by economic activities in other countries – it is a highly interconnected world.

Table 9 that follows below, summarises certain important features of CCMs and VCMs in a side-by-side format to help to understand their distinguishing features and their potential significance.

Table 9

Voluntary Carbon Market and Compliance Carbon Market – Comparing Features

Feature	Voluntary Carbon Market (VCM)	Compliance Carbon Market (CCM)
Commodity	Carbon Offsets – they are certificates awarded to represent the ‘product’ called the carbon offset that is created through ‘carbon sequestration’ which is a natural or artificial process by which carbon dioxide is removed from the atmosphere and held in solid or liquid form. Facilitated by project-based system.	Carbon Allowances – they are permits to emit carbon dioxide Facilitated by cap-and-trade system Carbon Credits – like carbon allowances are usually created by the government – they are a measurement unit to repay a business for investing in green projects or initiatives that reduce carbon emissions.
Regulation	Operate on a self-regulatory basis where participants have voluntary commitments to reduce CO ₂ emissions. There is no formal or direct regulation by governments or financial authorities. As the VCM is still a nascent there are discussions in most jurisdictions about how matters such as transparency and integrity need to be addressed.	Regulated according to a specific emissions trading system. CCMs are implemented by governments to achieve specific carbon emission reduction targets, usually to comply with the nation’s Nationally Determined Contributions (NDCs) under the Paris Agreement (2015) and more specifically making participation mandatory for ‘in scope’ entities.
Price	Determined in a specific project with chosen participants – as with all commodity markets prices are driven by supply and demand as well as specific project characteristics, buyer preferences and quality (so projects certified under rigorous standards for biodiversity, community and climate are likely to command higher prices.	Determined by the processes of the specific emissions trading system and international carbon markets. The carbon prices are determined by market forces of supply and demand for emissions allowances which are the permits to emit carbon dioxide – each ETS operates according to a cap on total emissions and allowances are traded.
Participants	Companies or similar business organisations that take the opportunity to work with other businesses or individuals motivated to offset their carbon emissions. In recent years (from 2021) active new entrants are seen to include oil and gas majors, hedge funds and banks.	Companies or businesses that are in the scope of a regulated emissions trading system – they are issued a certain number of carbon allowances or credits (the terms may be understood to mean the same thing) that are like certificates each granting permission to generate one ton of CO ₂ emissions.
Where traded	It is a voluntary market, there is no centralized market but project developers or companies can sell carbon offsets (certificates) to investors/buyers through brokers or retailers or digital platforms. There are also initiatives by very large entities such as Shell and BCG (2021) that show the strategic importance of major purchasers of carbon offsets in the VCM.	The marketplace is an organised regulated market set by cap-and-trade regulations at regional or state level. The marketplace for trading the regulated carbon allowances uses specific exchanges such as the Intercontinental Exchange (ICE) and the European Energy Exchange (EEX) that offer platforms for continuous trading so carbon allowances can be traded throughout each day.

Source: Based on ICE website information at

<https://www.ice.com>

and Carbon Credits .com website

<https://carboncredits.com/what-is-the-voluntary-carbon-market/>

and S&P Global at website

<https://www.spglobal.com/commodity-insights/en/news-research>

Carbon Allowances, Carbon Credits and Carbon Offsets

Some care is needed when discussing 'carbon allowances', 'carbon credits' and 'carbon offsets' – the term 'carbon credits' may be found to be used interchangeably with either 'carbon allowances' or 'carbon offsets' depending on the sources and any particular author's interpretation. The terms 'carbon crediting' and carbon credits are often used to describe the arrangements that are created by governments (or their delegated bodies) and are transacted in CCMs – but confusingly the terms such as 'offset system' or 'credit offset' are also used for certain government schemes in certain nations. Carbon offsets are also created by independent businesses (not governments) and are transacted in VCMs.

VIII The Development of CBAMs

According to World Bank data only 28 per cent of carbon emissions were covered by carbon pricing mechanisms in 2025 (World Bank, 2025a) . There are no doubt different views on whether this is closer to success or failure after the major international efforts over many years including about 30 'COPs'. With around 72 per cent of carbon emissions not covered by any form of carbon pricing system as implied from the World Bank (2025a) figures there may be a significant risk of 'carbon leakage' and other adverse implications for nations operating carbon pricing in a world that has not achieved global carbon pricing despite the laudable intentions.

Countries may decide to operate carbon pricing policies to deter 'free-riding' by jurisdictions that do not apply carbon pricing (MacKay et al., 2017). The idea of 'Climate Clubs' and the rationale for the imposition of carbon tariffs has been discussed by for example, Nordhaus (2017). The drive towards a global carbon pricing objective is a continuing process. At the time of writing only one carbon border adjustment mechanism (CBAM) is operating – the EU CBAM – and there has been some research into the reactions to this by the nations outside the EU, for example, the IETA found in a 2024 study that potential support for CBAMs varies across countries and income levels (IETA, 2024).

Other CBAMs are expected to be implemented in the next few years, for example, the UK is scheduled to introduce a CBAM in January 2027 (HM Treasury, 2025). The UK CBAM is likely to directly affect businesses that import GBP50,000 or more of specified goods from the aluminium, cement, fertiliser, hydrogen, and, iron and steel sectors over a 12-month period, while businesses that use such goods in their supply chain will be indirectly impacted, so this is expected to impact an estimated 10,000 businesses and will also impose on SMEs proportionately higher administrative costs relative to larger producers (HMRC, 2025). Another jurisdiction that may introduce a CBAM is Canada where 'Border Carbon Adjustments' (BCAs) are being explored by the Government of Canada to provide a tool to complement existing domestic carbon pricing instruments (Natural Resources Canada, 2025). It may not be surprising if Canada is considering entering bilateral agreements with the EU (IETA, 2024).

There are a number of significant jurisdictions in terms of the sizes of their economies and their GHGs including CO₂ emissions that are in the process of developing their own carbon pricing mechanisms and it may be that CBAMs will also be introduced in due course.

IX Some Controversies and Challenges in Carbon Pricing and Carbon Markets

The history of carbon pricing developments over about 50 years has shown that there are considerable controversies and challenges in achieving global carbon pricing that is consistent, far less uniform. Some of these are mentioned below.

1. The Kyoto Protocol (1997) discussed earlier, introduced the concept of national carbon emissions trading and the Clean Development Mechanism (CDM) but even though this appeared to be an achievement in reaching international cooperation the treaty has faced criticism as a case of institutional failure with inadequate design features such as too short a time frame for action, static targets for emission reductions, the form of calculation of emission reductions (CDMs), and multiple commitment periods perpetuating path-dependent structures (Rosen, 2015). It could be argued that a strength of the Kyoto Protocol was in helping to ensure that climate change remained on the international agenda but some of the major challenges and problems that confronted Kyoto have remained for The Paris Agreement (2015) such as problems of relevance and enforcement when significant carbon emitting countries are not members of the Paris Agreement as at 2025.
2. There are differing views on CBAMs – CBAMs are technically very important for the effectiveness of carbon pricing but there is considerable resistance due to international trade relations effects arising from concerns about economic impact on both exporting and importing countries.
3. Price transparency is an important feature to promote growth in carbon markets and there is a challenge especially in the VCM where the wide array of unique projects presents some complexity or even barriers for traders seeking accurate daily pricing for profit and loss and mark-to-market valuation purposes.
4. The VCM by its nature is subject to variations in carbon offset project quality and concerns over quality assurance so that potential buyers may be deterred if the carbon offsets are shown to have been oversold by subsequent research evidence.
5. Carbon pricing that includes carbon taxes and emissions trading systems can have regressive effects on possibly sizeable sections of a nation's population especially the lower income but also the middle income or 'middle wealthy' within the population and the theories of compensatory recycling of tax revenues or revenues from sales of carbon allowances may not always occur in practice.
6. After many years of international institutional discussions only 28 per cent of the world's carbon emissions by 2025 (World Bank, 2025a) were subject to carbon pricing systems.
7. Measuring carbon emissions and the effectiveness of carbon pricing systems is complex and prone to error and manipulation – there remain many questions as to how these problems can be addressed in the future.

X Conclusion

The paper has considered carbon pricing policies that have developed in response to global climate change concerns. There have been significant international institutional climate change policy drivers to harness a global effort to reduce adverse anthropogenic (human-induced) effects on the climate system. Following the Kyoto Protocol (1997) and then the Paris Agreement (2015) many countries have adopted carbon pricing policies with various mechanisms that in combination had achieved coverage of some 28 per cent. of global carbon emissions by 2025 according to the World Bank. There are both mandatory and voluntary carbon markets that are effectively specialised human-construct commodity markets. Carbon pricing has encouraged structured financial engineering based on measured carbon emissions in different jurisdictions. Further research will be necessary to assess whether the progress in carbon pricing and carbon markets is sufficiently effective in achieving positive outcomes at national and global levels. Governments may face increasing pressure to consider the balance between carbon pricing policies and promotion of carbon management technologies especially if the latter prove to be more effective and/or acceptable to businesses and the wider public. This paper has identified some controversies and challenges in the development of carbon pricing and has considered important issues to be addressed in the future.

Glossary

CBAM Carbon Border Adjustment Mechanism

CCMs Compliance Carbon Markets

CDM Clean Development Mechanism

CER Certified Emission Reduction (credits)

COP Conference of the Parties

EDGAR Emissions Database for Global Atmospheric Research

EEX European Energy Exchange

ERU Emission Reduction Units

ETS Emissions Trading System

EU ETS European Union Emissions Trading System

GHG Greenhouse Gas

ICE Intercontinental Exchange

ICC International Chamber of Commerce - an institution that has a stated objective to promote international trade, responsible business conduct and a global approach to regulation while also providing dispute resolution services – it reports that it is the institutional representative of more than 45 million companies in over 170 countries.

ICAP International Carbon Action Partnership – an international forum founded in 2007 for governments and public authorities that have implemented or are planning to implement emissions trading systems (ETSs).

IEA International Energy Agency – is an intergovernmental organization that has an expressed objective to work with governments and industry with a view to shaping a secure and sustainable energy future.

IETA International Emissions Trading Association – formed in June 1999, it is a non-profit organisation formed by a group of companies to promote a functioning international framework for trading greenhouse gas emissions reductions. IETA members seek to develop an emissions trading regime that results in real and verifiable greenhouse gas emissions reductions, balancing economic efficiency with environmental integrity and social equity. By April 2004, IETA had grown to have 88 international companies as its member and it actively cooperates with other international organisations including the World Bank,

the World Economic Forum and the World Business Council for Sustainable Development (see Select Committee on European Union Minutes of Evidence, 30 April 2004).

IPCC Intergovernmental Panel on Climate Change – formed in 1988, it is an intergovernmental body of the United Nations formed in 1988 and headquartered in Geneva, Switzerland, with 195 member states. Its parent organization is the WMO.

JRC (European Commission) Joint Research Centre is a department of the European Commission that provides independent, evidence-based knowledge and science, supporting EU policies to positively impact society.

NDCs Nationally Determined Contributions

REDD+ Reducing Emissions from Deforestation and forest Degradation in Developing Countries – there are carbon offsets from voluntary ‘Avoided Deforestation’ projects that are based on predictions of tree loss that would have occurred in the absence of the REDD+ scheme.

UK ETS United Kingdom Emissions Trading Scheme

UNFCCC United Nations Framework Convention on Climate Change – a multilateral treaty adopted in 1992 and entered into force in 1994 with the purpose of stabilizing greenhouse gas concentrations at a level that would prevent dangerous anthropogenic (human-induced) interference with the climate system.

VCM Voluntary Carbon Market

WMO World Meteorological Organization – formed in 1959, it is a specialized agency of the United Nations responsible for promoting international cooperation on atmospheric science, climatology, hydrology and geophysics.

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Appendix**Figure A – Conference to the Parties (COP) to the UNFCCC 1995 to 2025**

Conference	Year	Venue	Main Outcome
COP1	1995	Berlin, Germany	Agreed to begin process for individual country action.
COP2	1996	Geneva, Switzerland	Accepted scientific findings on CC of IPCC
COP3	1997	Kyoto, Japan	Kyoto Protocol : GHG emissions reduction obligation for ‘Annex 1’ countries plus ‘ Kyoto Mechanisms ’
COP4	1998	Buenos Aires, Argentina	Buenos Aires Plan of Action to implement Kyoto P.
COP5	1999	Bonn, Germany	165 parties considered path to finalizing Kyoto P.
COP6	2000	The Hague, Netherlands	Disagreements over US proposal for ‘ carbon sinks ’.
COP7	2001	Marrakech, Morocco	Marrakech Accords Established Adaptation Fund (AF).
COP8	2002	New Delhi, India	Call for technology transfer to minimize impact of CC
COP9	2003	Milan, Italy	Agreement to use AF to support developing countries.
COP10	2004	Buenos Aires, Argentina	Plan of Action: help developing countries cope with CC
COP11	2005	Montreal, Canada	Montreal Action Plan to extend Kyoto beyond 2012.
COP12	2006	Nairobi, Kenya	Agreed on procedures for AF for developing countries.
COP13	2007	Bali, Indonesia	Agreement on timeline for post-2012 negotiations.
COP14	2008	Poznan, Poland	Nations agreed on principles for a CC ‘AF’ fund
COP15	2009	Copenhagen, Denmark	Noted ‘Political Accord’ 25 parties incl USA and China
COP16	2010	Cancun, Mexico	Call for Green Climate Fund – USD 100 billion per year.
COP17	2011	Durban, South Africa	Green Climate Fund (GCF) framework adopted.
COP18	2012	Doha, Qatar	Doha Amendment to Kyoto Protocol – disagreements .
COP19	2013	Warsaw, Poland	Warsaw Framework for REDD-plus and Warsaw International Mechanism (WIM) for Loss and Damage.
COP20	2014	Lima, Peru	Negotiations for a global climate agreement.
COP21	2015	Paris, France	The Paris Agreement was to be the basis for participant nations’ CC reduction measures from 2020.
COP22	2016	Marrakech, Morocco	Focus on water-related sustainability with consideration of water scarcity and water cleanliness.
COP23	2017	Bonn, Germany	Launch of the Powering Past Coal Alliance (PPCA) a group of 185 nations, cities, regions and organisations.
COP24	2018	Katowice, Poland	There was agreement on rules to implement Paris Agreement including measurement and reporting.
COP25	2019	Madrid, Spain	Spain maintained sequence of COPs after Brazil then Chile withdrew from hosting the scheduled COP25.
Virtual	2020	Virtual due to Covid19	Aimed to maintain for CC events and dialogues.
COP26	2021	Glasgow, UK	Glasgow Climate Pact : Decisions in the collective effort to limit global temperature rise to 1.5 Degrees C
COP27	2022	Sharm el-Sheikh, Egypt	Five key ‘takeaways’ : establishing a dedicated fund for loss and damage; maintaining a clear intention to keep 1.5 Degrees C within reach; holding businesses and institutions to account; mobilizing more financial support for developing countries; making the pivot towards implementation.
COP28	2023	Dubai, UAE	Marked conclusion of the first ‘global stocktake’ of efforts to address climate change under Paris Agreement – call on governments to speed up transition from fossil fuels to renewables (wind/solar).
COP29	2024	Baku, Azerbaijan	Agreement to triple finance to developing countries from previous goal of USD 100 billion annually, to USD 300 billion annually by 2035
COP30	2025	Belem, Brazil	Due to take place from 10-21 November, 2025

Source: United Nations Climate Change: Past Conferences Overview. Accessed at:

<https://unfccc.int/process-and-meetings/conferences/past-conferences/past-conferences-overview>

Appendix

Figure B – Greenhouse Gases

GHG	Greenhouse Gas
CO2	Carbon Dioxide
H2O	Water Vapour
CH4	Methane
N2O	Nitrous Oxide
HFCs	hydrofluorocarbons
PFCs	perfluorocarbons
SF6	Sulphur Hexafluoride
NF3	Nitrogen Trifluoride

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